Learning Objective 6.4: I can implement an mPSK modulator and demodulator (BPSK, QPSK, 8PSK), recover carrier and symbol timing, and verify constellation behavior against theory.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Bits per symbol.** A telemetry radio sends 25,000 symbols per second with RRC pulses of roll-off $\alpha = 0.25$. Lesson 3 gives the E_b/N_0 each modulation needs for a bit error rate of 10^{-5}: 9.6 dB for BPSK and QPSK, and 13.0 dB for 8PSK.

   (a) Find the occupied bandwidth, and the bit rate with BPSK, QPSK, and 8PSK.

   B:

   bits:

   (b) The link budget delivers $E_s/N_0 = 15$ dB at the receiver. Find E_b/N_0 for each modulation, and say which ones reach a bit error rate of 10^{-5}.

   (c) In words, explain why this radio would use QPSK rather than BPSK, and what it would give up by moving to 8PSK.

2. **The Gray map.** The L15 transmitter numbers the QPSK points by *phase index* $i = 0, 1, 2, 3$, counterclockwise from the point at $45°$. The Gray label of index i is $i \oplus (i \gg 1)$, the index XORed with itself shifted right one bit.

   (a) Write the 2-bit Gray label of each index. Then find the indices the transmitter sends for the bits `01 11 10 00`.

   labels: ____

   indices: ____

   (b) Noise moves a symbol sent as index 1 into the region of index 2. How many bits are wrong with the Gray labels? How many would be wrong if index i simply carried the binary number i?

   Gray: ____

   binary: ____

   (c) In words, explain why the BER of a Gray-coded 8PSK link is close to its symbol error rate divided by 3.

3. **Differential coding by hand.** The L15 Differential Encoder sends $e_n = (e_{n-1} + d_n) \bmod M$, where d_n is the phase index from the Gray map and e_{n-1} starts at 0. The decoder outputs $(r_n - r_{n-1}) \bmod M$, where r_n is the received index and r_{n-1} also starts at 0. The link uses QPSK, and the data indices are $d = 2, 1, 1, 3, 0$.

   (a) Find the five indices the encoder sends.

   (b) The receiver's Costas Loop locks 180° off, so every received index is 2 more than the one sent, mod 4. Decode the five received indices and compare them with d.

   (c) Now the loop locks correctly, but noise turns the third received index from 0 into 1. Decode the five indices and count the wrong symbols.

   (d) In words, use parts (b) and (c) to explain what differential coding does for a link and what it does to the bit error rate.

4. **The frequency offset.** A frequency offset Δf turns the constellation by $360^\circ \times \Delta f / R_s$ every symbol. Lesson 15 measured the pull-in range of the QPSK Costas Loop at about 375 Hz for $R_s = 4{,}000$ symbols per second and an E_b/N_0 of 12 dB. The loop works once per symbol, so with the same loop settings its pull-in range scales with the symbol rate.

   (a) A 4 kbaud QPSK link has a 75 Hz offset. Find the rotation per symbol and the number of symbols in one full turn of the constellation.

   (b) Two radios use oscillators accurate to 2.5 ppm and transmit at 2.4 GHz. Find the largest frequency offset between them. Can a 4 kbaud QPSK Costas Loop lock to it? Can a 1 Mbaud one?

   4 kbaud:

   1 Mbaud:

   (c) In words, explain why an 8PSK Costas Loop pulls in from a smaller offset than a BPSK loop with the same bandwidth, and what the 4 kbaud receiver in part (b) needs before its Costas Loop.

5. **QPSK or 8PSK?** A link must carry 12 kb/s in a channel 6.25 kHz wide, with RRC pulses of roll-off $\alpha = 0.35$ and differential coding. Lesson 15 measured the E_b/N_0 the differential link needs for a bit error rate of 10^{-4}: 8.9 dB for QPSK and 12.4 dB for 8PSK.

   (a) Find the symbol rate and the occupied bandwidth with QPSK and with 8PSK. Which fits the channel?

   QPSK:

   8PSK:

   (b) The bit rate is the same either way. How much more received power does the 8PSK link need for a bit error rate of 10^{-4}?

   (c) In words, state which modulation you would choose and what the link budget of Lesson 2 has to show before you commit to it.